

RE: How I fixed Arable Labs, Inc. Mark II micro weather station rainfall measurements to reduce errors and stop major customer loss

Arable labs suffered from a technical problem involving its micro-weather station Mark II. The device used a microphone which captured acoustic energy to measure rainfall. The device ran into some major production problems with its failure to minimize false positives (no rainfall reported by device, but actual rainfall) and false negatives (rainfall reported, no actual rainfall). The problem was a severe one, causing > \$1M revenue customers for instance in Nebraska, to discontinue product use due to poor technical performance.

Objective: to minimize false positives and negatives (Type I, II errors) and fix the device.

Solution: the solution (found and implemented exclusively by Michael Yang and coded by him into the v2 firmware for Mark II) was to break apart the audio signal into 128 discrete FFT bins scaled down to 32 bins e.g.:

Rainfall sounds from low to high energy:

FFT output energies:

[4] [3] [1] [1] [0] [0] [0] [0] [0] [1] [0]

Then apply a least square minimization between our actual Hawaii rainfall disdrometer (expensive rainfall gauge) and the individual FFT bins obtained from breaking apart the sound signal.

Results: A higher-dimensionality feature vector was necessary to minimize false positives and negatives and retain customer relationships. A least square minimization was used to maximize likelihood, and was successful in *retaining customers and fixing the core rainfall technology of Arable Labs, Inc.*